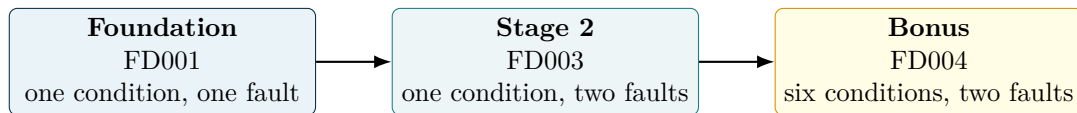


Jet Engine Hospital

Leakage-Safe Multi-Task Predictive Maintenance
on NASA C-MAPSS



Student: Partow Roshani
Instructor: Dr. Farahani
Course: Machine Learning Capstone
Project: NASA C-MAPSS Turbofan Prognostics

Abstract

This project develops an auditable predictive-maintenance system for NASA C-MAPSS turbofan engines. It jointly estimates Remaining Useful Life (RUL), predicts failure within 10, 20, and 30 cycles, detects abnormal behavior without failure labels, quantifies uncertainty, and converts the evidence into a maintenance recommendation. The study follows the required progression: FD001 is the foundation benchmark, FD003 introduces multiple fault modes, and FD004 combines six operating conditions with two fault modes as a bonus extension. All partitions are created by engine identifier before preprocessing, feature construction, calibration, threshold selection, anomaly fitting, or uncertainty estimation. On official terminal-row benchmarks, the locked RUL pipelines achieve MAE values of 20.30, 35.70, and 25.63 cycles for FD001, FD003, and FD004. The final FD004 policy reaches 92.68% critical recall, 89.96% exact-state accuracy, and a 0.0485% under-escalation rate. The complete model stack is serialized and used by a unified Streamlit dashboard for FD001, FD003, and FD004.

1 Problem Framing and System Design

For engine i at cycle t , the observed vector $\mathbf{x}_{i,t}$ contains three operating settings and 21 sensor measurements. The system produces three complementary outputs:

$$\widehat{\text{RUL}}_{i,t} \approx T_i - t, \quad (1)$$

$$\hat{p}_{i,t}^{(h)} = P(\text{RUL}_{i,t} \leq h \mid \mathbf{x}_{i,1:t}), \quad h \in \{10, 20, 30\}, \quad (2)$$

$$a_{i,t} = \text{unsupervised anomaly severity}. \quad (3)$$

The outputs are combined into the operational states NORMAL, WATCH, WARNING, and CRITICAL, which map to CONTINUE, INSPECT, or STOP. The decision remains traceable to the RUL interval, calibrated horizon risks, anomaly evidence, persistence, and the active policy rule. Anomaly evidence is capped at Watch and cannot independently Stop.

2 Data Audit and Leakage Controls

The supplied C-MAPSS files are whitespace-separated run-to-failure multivariate time series with 26 columns: engine ID, cycle, three operating settings, and 21 sensors. Training trajectories terminate at failure. Official test trajectories stop earlier and provide one terminal RUL value per engine. The raw-data audit checks engine counts, missing and non-finite values, duplicated (*engine_id, cycle*) keys, cycle continuity, sequence lengths, sensor variance, and horizon class balance.

Table 1: Datasets and engine-level partitions used by the final notebook.

Subset	Official train	Official test	Train/Val/Test	Conditions/Faults
FD001	100	100	70/15/15	1/1
FD003	100	100	70/15/15	1/2
FD004	249	248	174/37/38	6/2

The supplied FD004 files contain 249 training engines and 248 test engines; these counts are the reverse of the row printed in the project brief. The notebook uses the actual file contents and verifies count alignment before labeling.

The core anti-leakage rule is *split engines first*. Every temporal window stays inside one engine and one partition. Scalers, near-constant filtering, operating-regime clustering, feature selection, PCA, calibration, conformal residuals, anomaly thresholds, persistence settings, and policy thresholds are fitted only on their permitted training or validation engines. Explicit causality tests alter future rows after cycle t and confirm that features at t do not change. Centered rolling windows, whole-engine normalization, and final-cycle-derived predictors are excluded.

3 Feature Engineering and Model Families

FD001 retains 15 informative sensors and expands them into 138 causal model features. Features include current values, first differences, rolling means, rolling standard deviations, and local trend estimates. FD003 follows the same protocol to keep Stage 2 comparable. FD004 retains 17 paired sensors, creates 153 temporal sensor features, and reaches 164 inputs after condition context is added.

FD004 requires condition awareness. A six-cluster K-means model is fitted only on modeling-training operating settings and then frozen. Sensor values are represented through global scaling and operating-condition-normalized residuals; the selected representation combines condition-aware temporal features with explicit regime indicators.

Regression comparisons include Ridge, polynomial Ridge, Random Forest, Gradient Boosting, and explicit temporal-window models. FD004 additionally compares constrained Extra Trees and histogram gradient boosting. Classification compares calibrated Logistic Regression with nonlinear histogram gradient boosting. The anomaly benchmark uses Isolation Forest, Local Outlier Factor with `novelty=True`, One-Class SVM, and PCA reconstruction error, with score direction normalized so larger values always mean more abnormal.

4 Evaluation Protocol and Required Evidence

Model selection is performed on validation engines, while the internal test is touched only after the pipeline is locked. The official NASA test is used once for final terminal-row reporting. One master comparison keeps

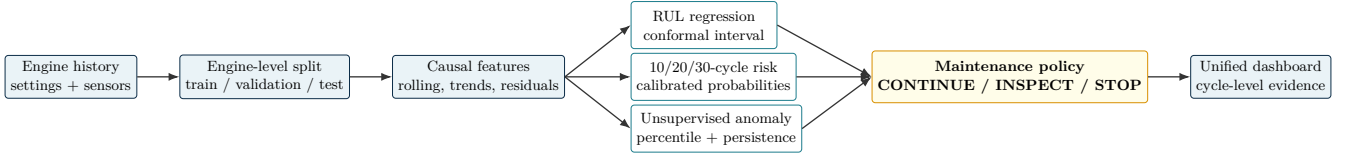


Figure 1: End-to-end architecture. Learned transformations and thresholds are fitted without using held-out test engines.

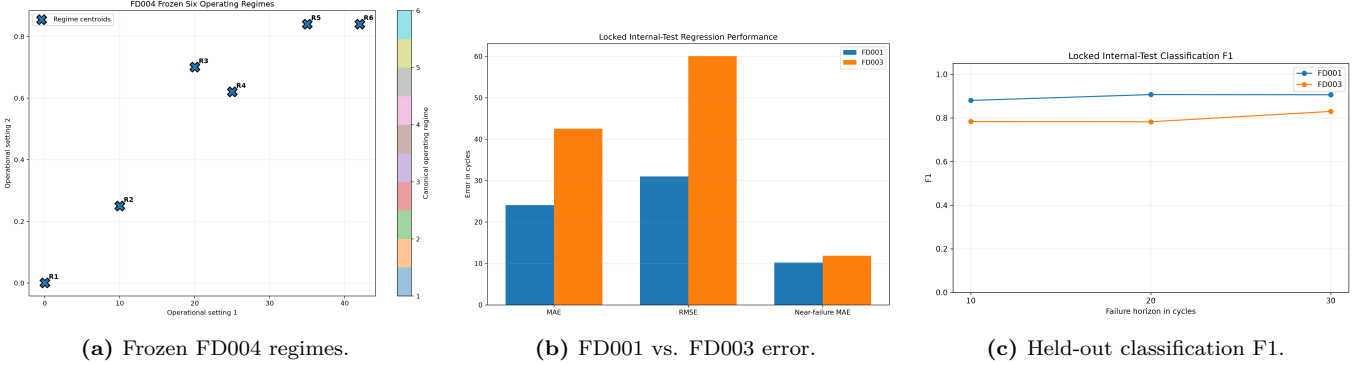


Figure 2: Condition structure and the main Stage 1/Stage 2 performance shift.

the evaluation units consistent across stages. Engine-level bootstrap resampling, rather than row-level bootstrap, is used for 95% confidence intervals so long trajectories do not dominate uncertainty estimates.

Table 2: Evaluation contract used throughout the project.

Task	Required evidence in the final notebook/report
RUL regression	MAE, RMSE, R^2 , NASA score, regional errors, residuals, engine traces, bootstrap CI, interval coverage/width
Failure risk	PR-AUC, precision, recall, F1, Brier score, calibration, confusion matrices, validation-selected thresholds
Anomaly	Four-detector comparison, normalized score direction, threshold sensitivity, false-alert and near-failure behavior
Early warning	First persistent alert, lead time, miss rate, late delay, early burden, asymmetric cost
Policy	State/action confusion, under- and over-escalation, critical and warning recall, trigger traceability

The notebook also reports model size and timing where relevant. Comparisons use the same engine IDs within a dataset, and the Stage 2 protocol remains fixed so that changes in performance reflect fault complexity rather than a new evaluation convention.

5 FD001 Foundation and FD003 Stage 2

FD001 is the controlled foundation with one condition and one fault mode. Degree-2 Polynomial Ridge with $\alpha = 0.01$ is selected on validation engines. Official terminal-row performance is MAE 20.304, RMSE 26.438, $R^2 = 0.595$, NASA score 4,421.83, and near-failure MAE 13.812 cycles. Its internal-test classification PR-AUC values are 0.954, 0.972, and 0.978 for the 10-, 20-, and 30-cycle horizons.

FD003 preserves one condition but adds two fault modes. The split, causal feature, calibration, uncertainty, anomaly, and policy conventions remain fixed. Official terminal regression degrades to MAE 35.703, RMSE 46.599, $R^2 = -0.267$, NASA score 18,224,484, and near-failure MAE 16.030. Despite this regression shift, the horizon classifiers remain safety-sensitive: internal-test recall is

0.988, 1.000, and 0.998. The central Stage 2 finding is therefore that multi-fault heterogeneity can break point RUL generalization while calibrated near-term alerting remains useful.

6 FD004 Combined Challenge

FD004 combines six operating conditions and two fault modes. Under Ridge regression, the best condition-aware representation reduces validation MAE from 37.394 for the global current representation to 37.123 cycles. The model-family comparison selects constrained Extra Trees, which achieves validation MAE 30.891, RMSE 39.976, $R^2 = 0.771$, near-failure MAE 9.056, 19.43 seconds fit time, and 0.129 seconds prediction time for 8,891 validation rows. This corresponds to approximately 14.5 ms per 1,000 rows in the recorded batch test.

A transfer stress test shows that a source-only FD003 current-cycle model is not suitable for FD004 without adaptation; the dedicated FD004 pipeline is therefore trained with frozen condition-aware preprocessing. The locked official terminal evaluation obtains MAE 25.631 cycles, RMSE 34.666, $R^2 = 0.596$, NASA score 261,491.54, mean bias 1.548, and near-failure MAE 9.120. Engine-level bootstrap 95% confidence intervals are [23.142, 28.468] for MAE and [31.431, 38.313] for RMSE.

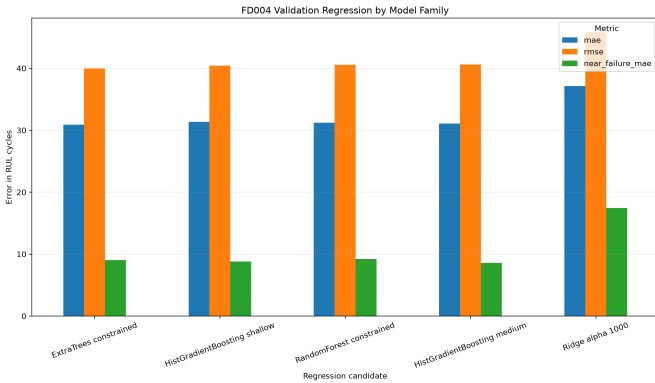
Row-level official MAE is 51.408 cycles, compared with 25.631 at the terminal row. Region diagnostics explain this gap: early-life MAE is 61.071, mid-life MAE is 28.909, and near-failure MAE is 9.928. The dashboard therefore presents uncertainty and risk jointly rather than treating a single early-life point estimate as definitive.

7 Failure Risk, Anomaly Detection, and Early Warning

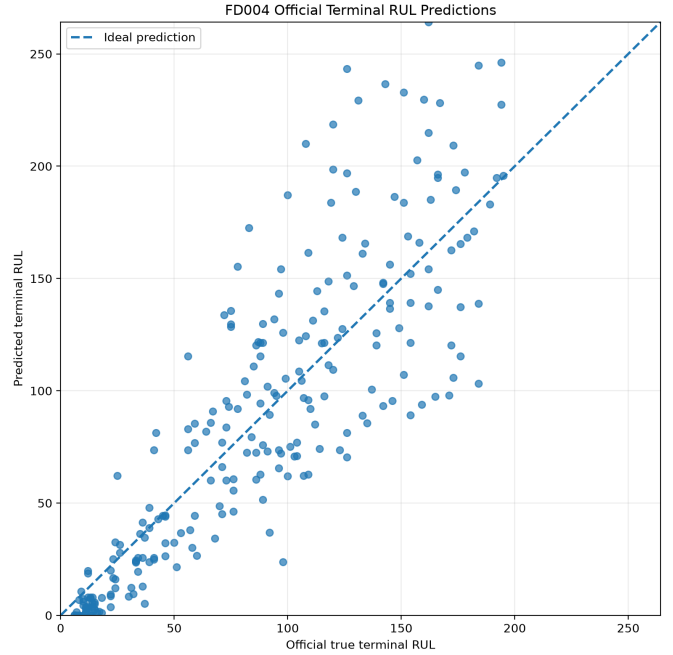
The deployed FD004 classifier uses separate Logistic Regression models for the three horizons with isotonic calibration and a locked threshold of 0.03. A nonlinear Hist-

Table 3: Official terminal-row RUL regression results.

Dataset	MAE	RMSE	R^2	NASA score	Near-failure MAE
FD001	20.304	26.438	0.595	4,421.83	13.812
FD003	35.703	46.599	-0.267	18,224,484.24	16.030
FD004	25.631	34.666	0.596	261,491.54	9.120



(a) Validation comparison of FD004 regression families.



(b) Official FD004 terminal RUL predictions.

Figure 3: Model selection and one-time official evaluation for the combined challenge.

GradientBoosting comparator is retained as a diagnostic baseline, but the calibrated linear models are deployed because they are interpretable, calibrated, and already satisfy the validation safety contract. On the internal test, PR-AUC is 0.900, 0.933, and 0.934; recall is 1.000, 1.000, and 0.993.

Table 4: Held-out internal-test failure classification.

Dataset	h	PR-AUC	Precision	Recall	F1
FD001	10	0.954	0.848	0.915	0.880
	20	0.972	0.871	0.946	0.907
	30	0.978	0.859	0.959	0.907
FD003	10	0.962	0.649	0.988	0.784
	20	0.957	0.643	1.000	0.783
	30	0.956	0.711	0.998	0.830
FD004	10	0.900	0.553	1.000	0.712
	20	0.933	0.559	1.000	0.717
	30	0.934	0.580	0.993	0.732

On official FD004 trajectory rows, the 10-/20-/30-cycle PR-AUC values are 0.3045, 0.6778, and 0.7650. Precision is 0.2099, 0.3625, and 0.3680; recall is 0.9268, 0.9759, and 0.9745; F1 is 0.3423, 0.5286, and 0.5343. Brier scores remain low at 0.0010, 0.0040, and 0.0077. The higher-horizon models provide earlier action time but create more false-positive burden.

Table 5: FD004 validation benchmark for four anomaly detectors.

Detector	PR-AUC	ROC-AUC	Prec.	Recall	F1
Isolation Forest	0.861	0.974	0.640	0.903	0.749
LOF	0.841	0.968	0.521	0.953	0.673
One-Class SVM	0.816	0.971	0.449	0.983	0.616
PCA reconstruction	0.749	0.947	0.596	0.827	0.692

Isolation Forest is selected. On official trajectories it obtains PR-AUC 0.3982, precision 0.2260, recall 0.8866, F1 0.3602, an early-life alert rate of 2.85%, and a near-failure alert rate of 88.66%. Because precision drops under official distribution shift, anomaly evidence remains Watch-level only.

Persistent warning timing is measured per held-out engine. On 38 FD004 internal-test engines, no horizon misses an engine before failure. Median lead times are 17.5, 34, and 53 cycles. Mean lead times are 20.68, 41.58, and 70.95 cycles, while mean asymmetric cost per engine rises from 10.68 to 21.58 and 40.95 because longer horizons increase early-warning burden.

8 Uncertainty and Unified Maintenance Policy

FD004 uses engine-grouped conformal residuals. The locked global interval reaches 99.78% coverage on validation and 92.41% on the internal test; near-failure internal-test coverage is 100%. On official terminal rows, coverage is 99.60% with average width 181.00 cycles. On all official trajectory rows, coverage is 88.76% with average width 212.76 cycles. The intervals are deliberately treated as advisory evidence, especially early in life.

The unified FD004 policy applies immediate escalation and requires two consecutive lower-state cycles for de-escalation. On official trajectories, exact-state accuracy is 0.899573, under-escalation is 0.000485, over-escalation is

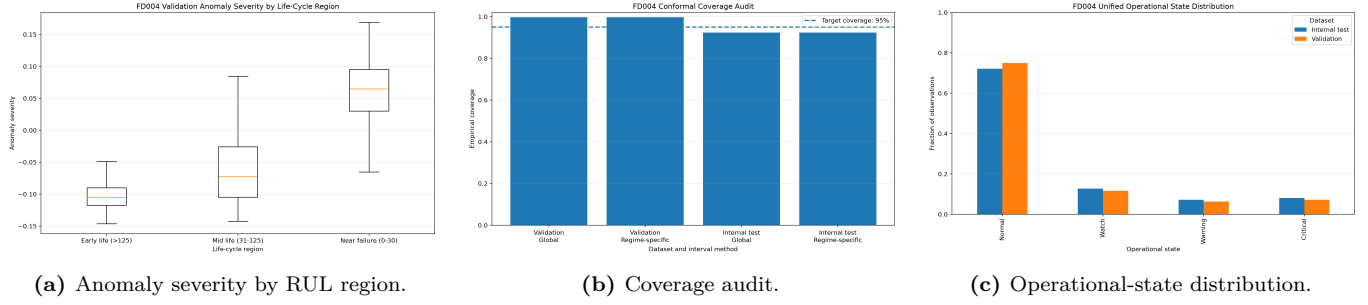


Figure 4: Unsupervised evidence, uncertainty, and policy behavior.

0.099942, severe under-escalation is zero, critical recall is 0.926829, warning-or-higher recall is 0.983957, watch-or-higher recall is 0.987269, and normal specificity is 0.908798. The conservative policy therefore minimizes safety-critical under-escalation at the cost of extra inspections.

9 Ablations and Error Analysis

The required ablations confirm which components matter and where complexity is unnecessary.

- **Operating settings on FD001:** removing the two varying settings changes internal-test MAE from 24.092 to 24.016 cycles, showing that these settings add little under a single operating condition.
- **Temporal representation on FD003:** temporal Ridge improves validation MAE from 51.619 to 48.727, RMSE from 66.479 to 64.448, and near-failure MAE from 18.288 to 14.873 compared with current-cycle Ridge.
- **Probability calibration on FD003:** calibration reduces Brier score by 0.00175, 0.00432, and 0.00445 for the 10-, 20-, and 30-cycle models.
- **Persistence on FD003:** a 2-of-3 rule reduces false positives from 175 to 161 at 20 cycles and from 189 to 172 at 30 cycles while preserving near-perfect recall.

Per-engine and regional diagnostics show that fleet averages hide catastrophic FD003 regression errors and that FD004 early-life estimates dominate row-level error. The report therefore distinguishes official terminal evaluation, all-row evaluation, regional MAE, per-engine traces, bootstrap intervals, and policy confusion rather than relying on one headline metric.

10 Dashboard Logic, Deployment, and Reproducibility

The submitted Streamlit application loads the exported artifacts and supports FD001, FD003, and FD004. Users upload a chronological engine trajectory, choose an engine and a cycle, and inspect: sensor history; predicted RUL and conformal bounds; calibrated 10-/20-/30-cycle risk trajectories with locked thresholds; anomaly evidence and threshold; recent persistence counts; the selected action and trigger reason; and model/dataset metadata. Status colors are green for CONTINUE, amber for INSPECT, and red for STOP.

Table 6: Dashboard evidence contract implemented in `streamlit_app.py`.

Control/display	Implemented behavior
Dataset and upload	Select FD001/FD003/FD004 and upload a chronological engine trajectory
Engine selector	Select one engine from a multi-engine file without losing the loaded results
Cycle selector	Select an observed cycle and draw the corresponding vertical marker on trajectories
RUL evidence	Point estimate, lower/upper conformal bounds, and full RUL timeline
Risk evidence	10/20/30 probabilities, locked thresholds, threshold margins, direct/persistent alerts
Anomaly evidence	Severity/percentile, locked threshold, margin, recent persistence count
Decision output	Colored status, CONTINUE/INSPECT/STOP action, trigger reason, confidence/disagreement evidence
Metadata	Dataset, artifact version, creation time, training engines, model families, policy settings

Table 7: Final reproducibility and delivery audit.

Requirement	Status
Engine-disjoint saved splits	PASS
Causal feature tests	PASS
10/20/30 calibrated risks	PASS
Four anomaly detectors	PASS
Conformal uncertainty and engine bootstrap CIs	PASS
Persistent warning and cost analysis	PASS
Raw-to-artifact prediction parity	PASS
Unified FD001/FD003/FD004 Streamlit app	PASS
Hugging Face static access page embedding live Streamlit app	PASS
FD004 capstone compliance checks	13/13

Table 8: Frozen deployment components.

Component	Locked contract
FD001 regression	Degree-2 Polynomial Ridge, $\alpha = 0.01$
FD003 stage	Multi-fault pipeline with the FD001-compatible protocol
FD004 regression	Condition-aware temporal representation, constrained Extra Trees, 14-cycle safety offset
Classification	Separate calibrated Logistic Regression models for 10/20/30 cycles; monotonic probabilities
Anomaly	Isolation Forest fitted on healthy training regions; percentile-based score
Uncertainty	Engine-grouped conformal residual quantile with reported coverage and width
Policy	Immediate escalation; two-cycle confirmed de-escalation; anomaly capped at Watch

The FD004 serialized artifact is 63.87 MiB and contains preprocessing objects, fitted models, calibrators, thresholds, conformal parameters, policy rules, and metadata. The public operational backend is the Streamlit Community Cloud application. The submitted Hugging Face Space is a public Static Space that embeds the same live Streamlit dashboard through `index.html`. The Space therefore provides a direct public access layer while Python inference is executed by the linked Streamlit deployment. This architecture is stated explicitly so that

the report matches the submitted files and does not claim native Python execution inside Hugging Face.

Hugging Face landing page:

<https://huggingface.co/spaces/PartowRoshani/jet-engine-hospital>

Live Streamlit application:

<https://jet-engine-hospital-cmapss-egywtbmnfn.czliqitpnc4.streamlit.app/>

Repository:

<https://github.com/PartowRoshani/jet-engine-hospital-cmapss>

The project package includes one unified executable notebook, saved splits, source modules, tests, deployment artifacts, application code, figures, and result tables. The supplied automated test suite reports 64 passing tests. Frozen model metadata and manifest files support parity checks and repeatable inference.

Reproduction workflow

The locked environment specifies `scikit-learn==1.9.0`, `pandas==3.0.3`, and `streamlit==1.59.2`. The following commands reproduce the main checks from the project root:

```
python -m venv .venv
.venv/Scripts/activate (Windows)
pip install -r requirements-lock.txt
python -m pytest -q
streamlit run
    huggingface_space/streamlit_app.py
```

The unified notebook is `jet_engine_hospital_capstone_main_final.ipynb`. A clean top-to-bottom execution rebuilds the result tables, figures, and exported contracts. Because the notebook retrains and evaluates multiple models across three datasets, it is computationally heavier than the dashboard smoke test.

11 Limitations and Conclusion

Three limitations remain. First, early-life RUL is intrinsically uncertain and drives much of the all-row error. Second, FD003 shows that multiple fault modes can severely damage point-regression generalization even when near-term classification remains reliable. Third, the FD004 anomaly detector experiences a validation-to-official precision drop, so it is intentionally prevented from causing Stop by itself. In addition, the Hugging Face submission is a static wrapper rather than a native Python Space.

Despite these limitations, the machine-learning pipeline fulfills the substantive capstone objective: leakage-safe preprocessing, causal temporal features, classical and non-linear comparisons, separate calibrated risks for three horizons, four unsupervised detectors, uncertainty coverage, engine-level bootstrap confidence intervals, asymmetric early-warning analysis, an auditable maintenance policy, and a unified interactive dashboard. The main engineering conclusion is that no single score is sufficient; reliable maintenance support requires visible agreement and disagreement among RUL, near-term risk, anomaly evidence, and uncertainty.

Appendix: Final Locked Scorecards

Table 9: FD004 official regression and uncertainty with engine-level 95% confidence intervals.

Metric	Estimate	95% CI lower	95% CI upper
Terminal MAE	25.631	23.142	28.468
Terminal RMSE	34.666	31.431	38.313
Terminal R^2	0.596	0.476	0.683
NASA score	261,491.5	75,009.5	525,837.3
Conformal coverage	0.996	0.988	1.000
Average interval width	181.004	175.840	186.380

Table 10: FD004 official classification scorecard.

h	Positive rows	PR-AUC	Brier	Precision	Recall	F1	Threshold
10	41	0.3045	0.0010	0.2099	0.9268	0.3423	0.03
20	374	0.6778	0.0040	0.3625	0.9759	0.5286	0.03
30	864	0.7650	0.0077	0.3680	0.9745	0.5343	0.03

Table 11: FD004 persistent early-warning timing on 38 internal-test engines.

h	Misses	Median lead	Mean lead	Median early burden	Mean cost/engine
10	0	17.5	20.68		10.68
20	0	34.0	41.58		21.58
30	0	53.0	70.95		40.95

Table 12: Official FD004 maintenance-policy performance.

Metric	Value	Metric	Value
Exact state accuracy	0.899573	Critical recall	0.926829
Under-escalation rate	0.000485	Warning-or-higher recall	0.983957
Over-escalation rate	0.099942	Watch-or-higher recall	0.987269
Severe under-escalation	0.000000	Normal specificity	0.908798
Mean state transitions	3.814516	Maximum transitions	25

Additional FD004 Official Diagnostics

Table 13: Engine-bootstrap 95% confidence intervals for selected official trajectory metrics.

Metric	Estimate	95% CI lower	95% CI upper
H10 PR-AUC	0.3045	0.1570	0.6647
H20 PR-AUC	0.6778	0.5056	0.8524
H30 PR-AUC	0.7650	0.6476	0.8837
Anomaly PR-AUC	0.3982	0.2692	0.5664
Anomaly recall	0.8866	0.8209	0.9458
Policy exact-state accuracy	0.8996	0.8842	0.9146
Policy under-escalation	0.000485	0.000122	0.000971
Policy over-escalation	0.09994	0.08500	0.11533

Table 14: Final deployment resources and parity checks.

Item	Final value
FD004 artifact size	63.87 MiB
FD004 validation prediction time	0.129 s for 8,891 rows
Approximate batch latency	14.5 ms per 1,000 rows
FD004 compliance audit	13/13 PASS
Automated project tests	64 PASS
Raw-to-artifact inference parity	PASS
Unified app datasets	FD001, FD003, FD004

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